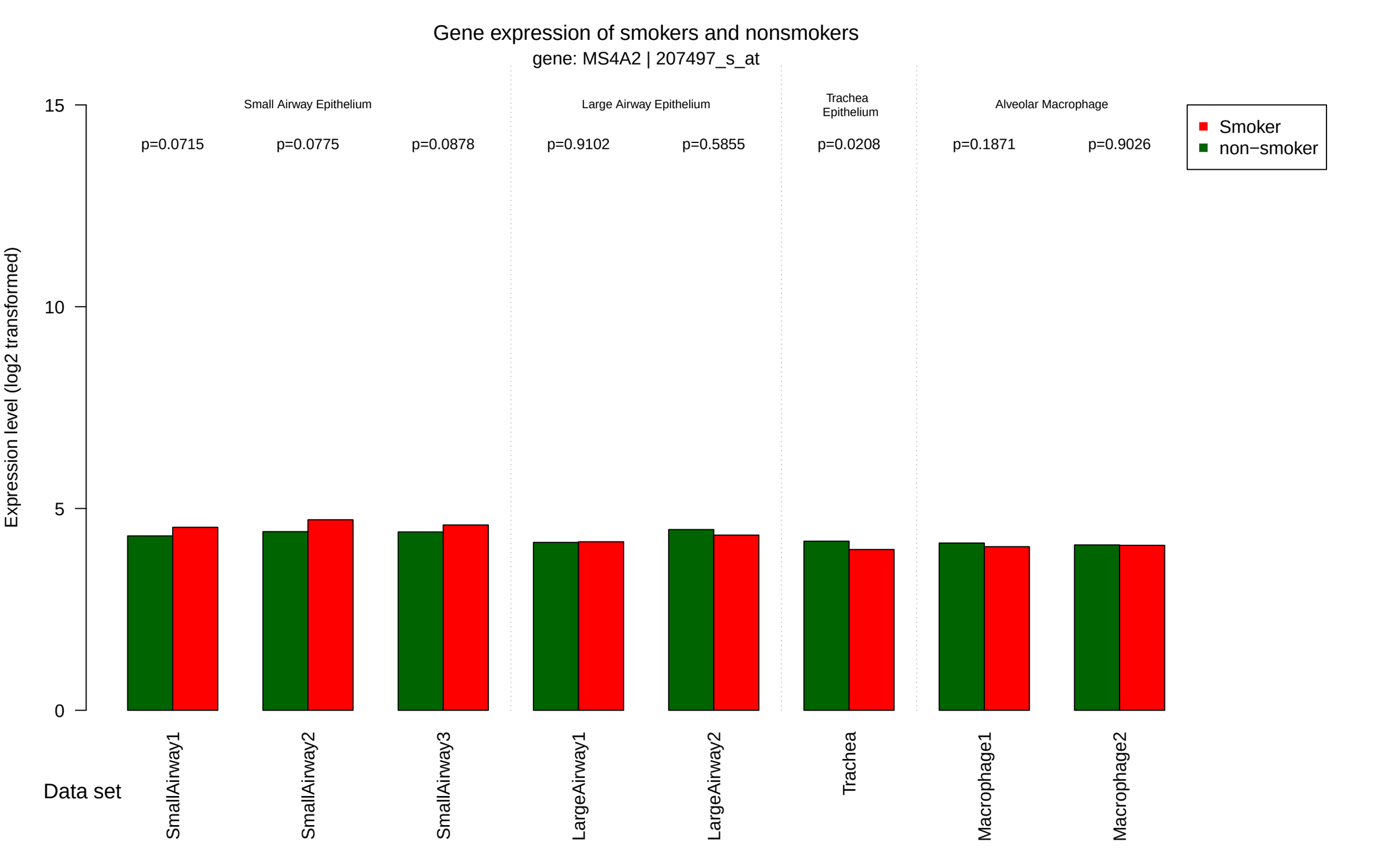
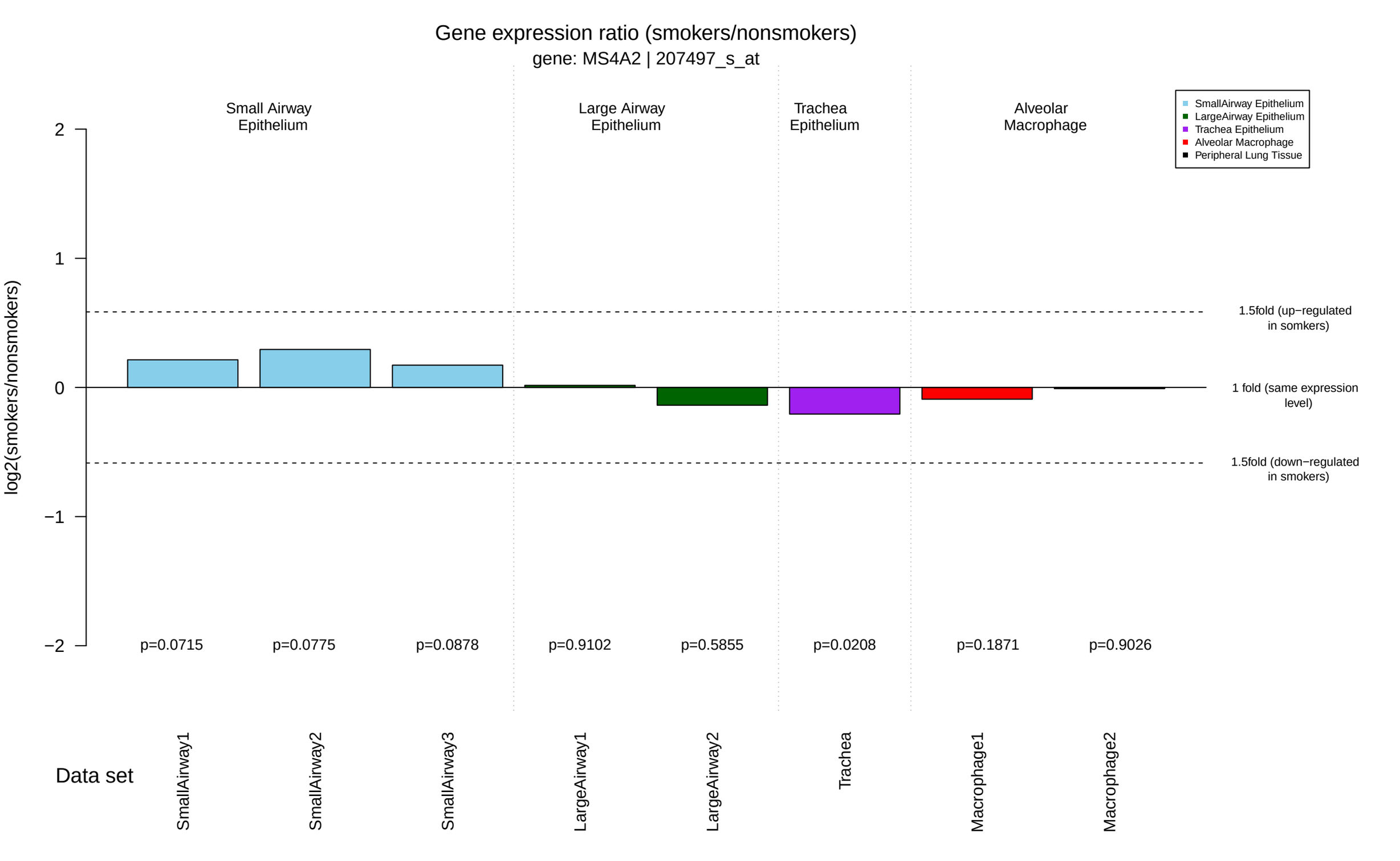
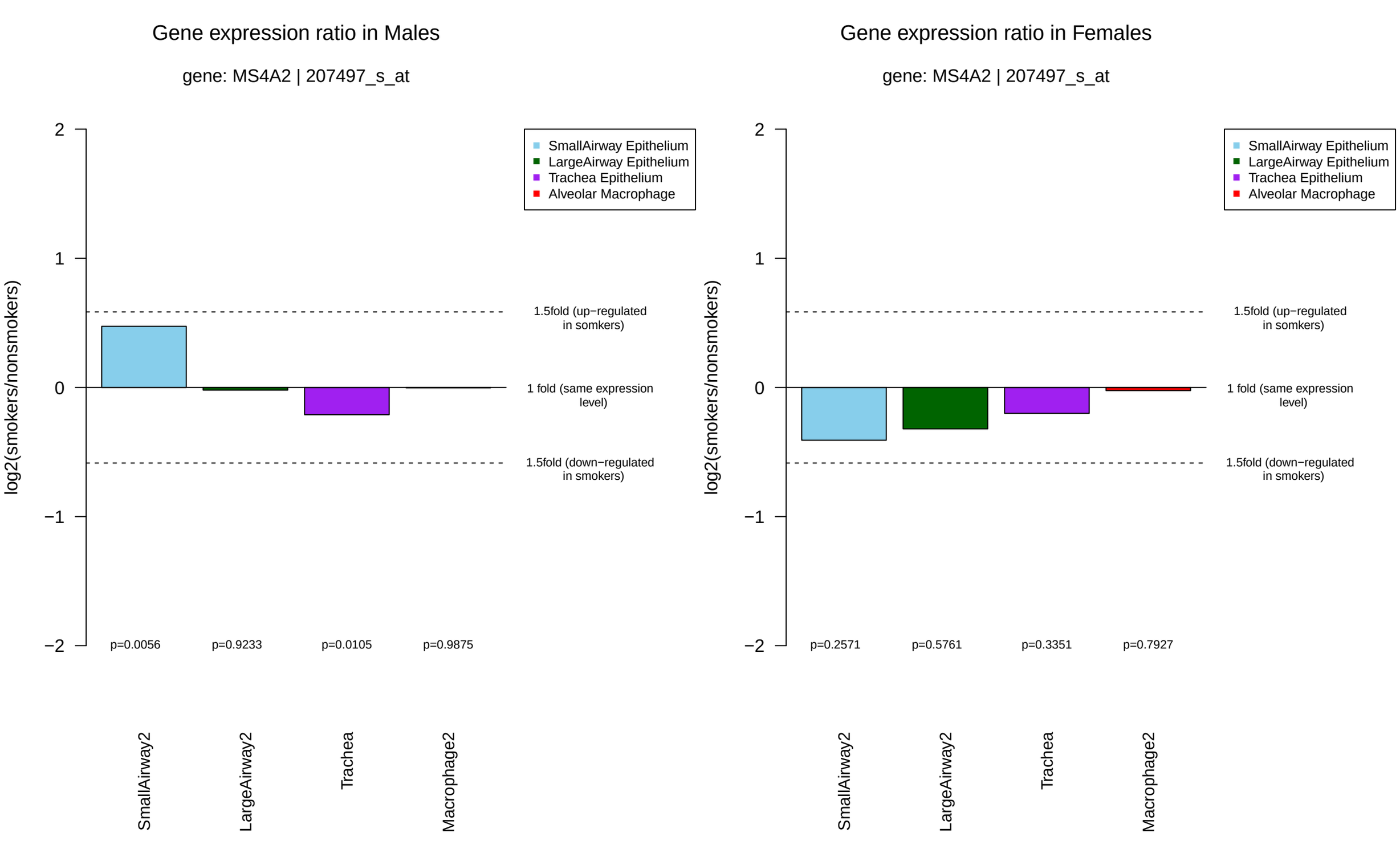
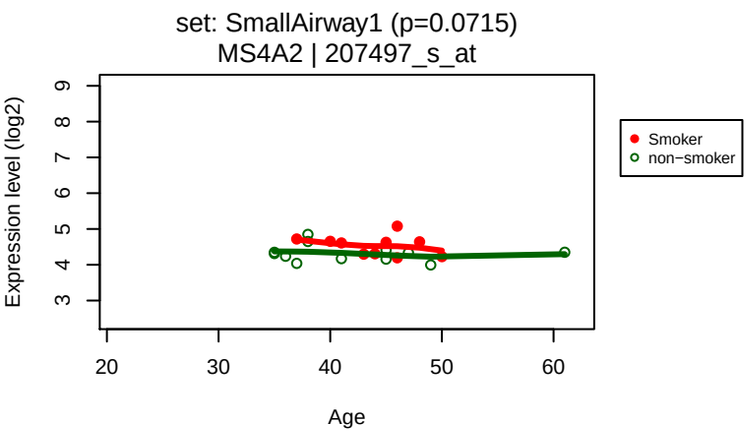
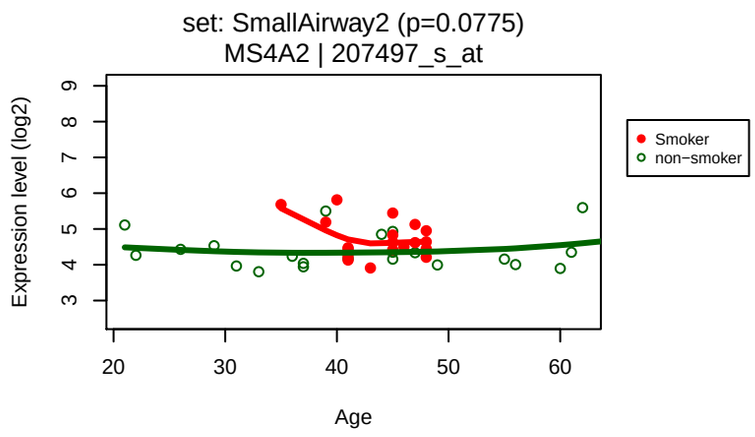
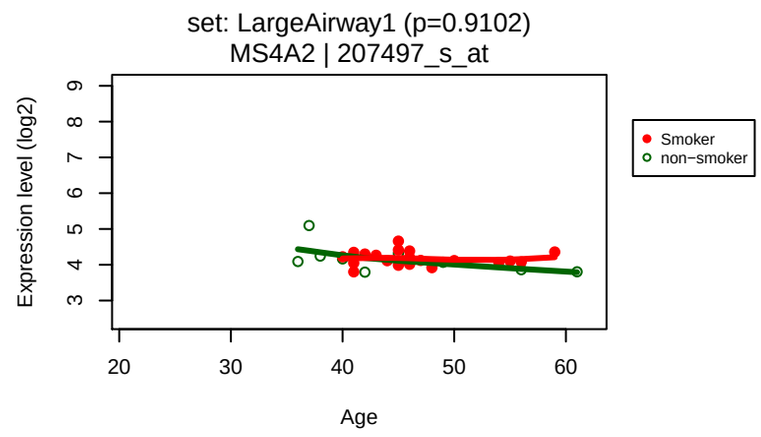

Gene expression of smokers and nonsmokers
gene: MS4A2 | 207497_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0715
p=0.0775
p=0.0878
p=0.9102
p=0.5855
p=0.0208
p=0.1871
p=0.9026
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

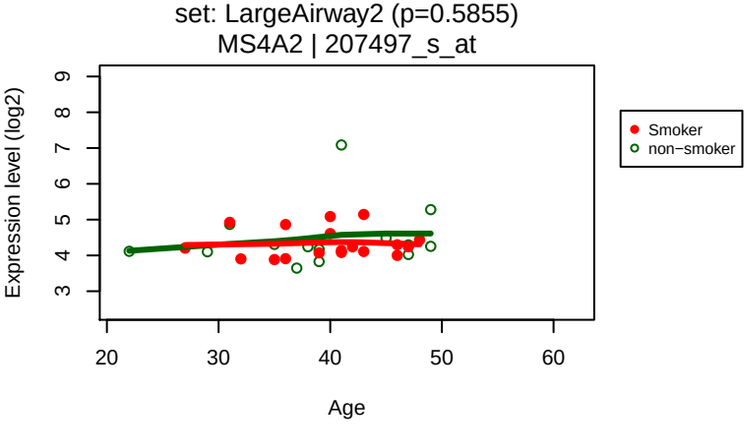

Gene expression ratio (smokers/nonsmokers)
gene: MS4A2 | 207497_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0715
p=0.0775
p=0.0878
p=0.9102
p=0.5855
p=0.0208
p=0.1871
p=0.9026
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

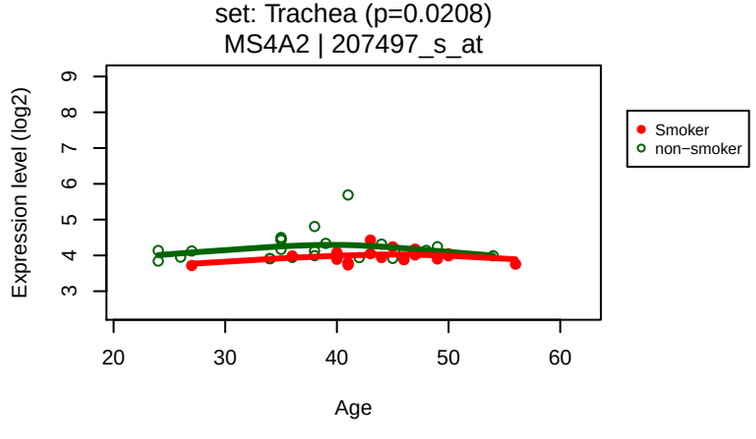

Gene expression ratio in Males
gene: MS4A2 | 207497_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0056
p=0.9233
p=0.0105
p=0.9875
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MS4A2 | 207497_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2571
p=0.5761
p=0.3351
p=0.7927
SmallAirway2
LargeAirway2
Trachea
Macrophage2

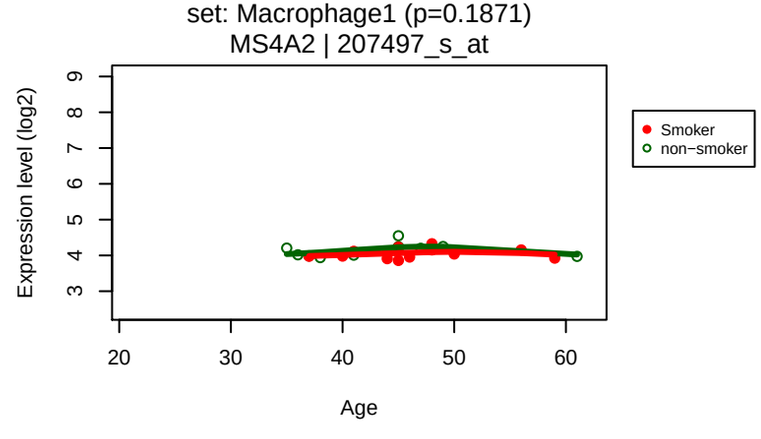

set: SmallAirway1 (p=0.0715)
MS4A2 | 207497_s_at
Expression level (log2)
Age
Smoker
non−smoker

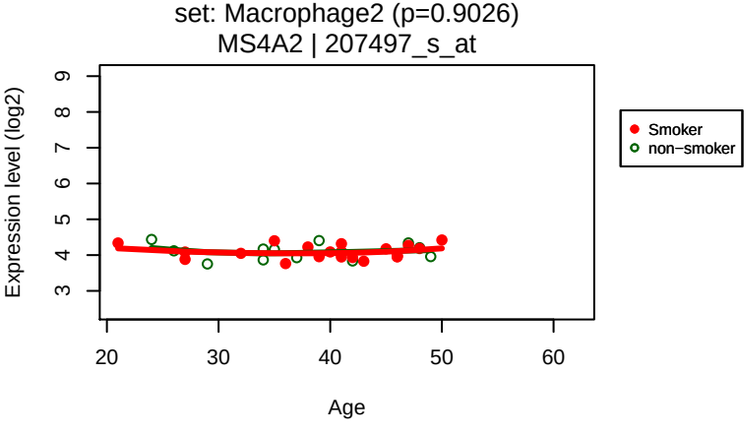

set: SmallAirway2 (p=0.0775)
MS4A2 | 207497_s_at
Expression level (log2)
Age
Smoker
non−smoker

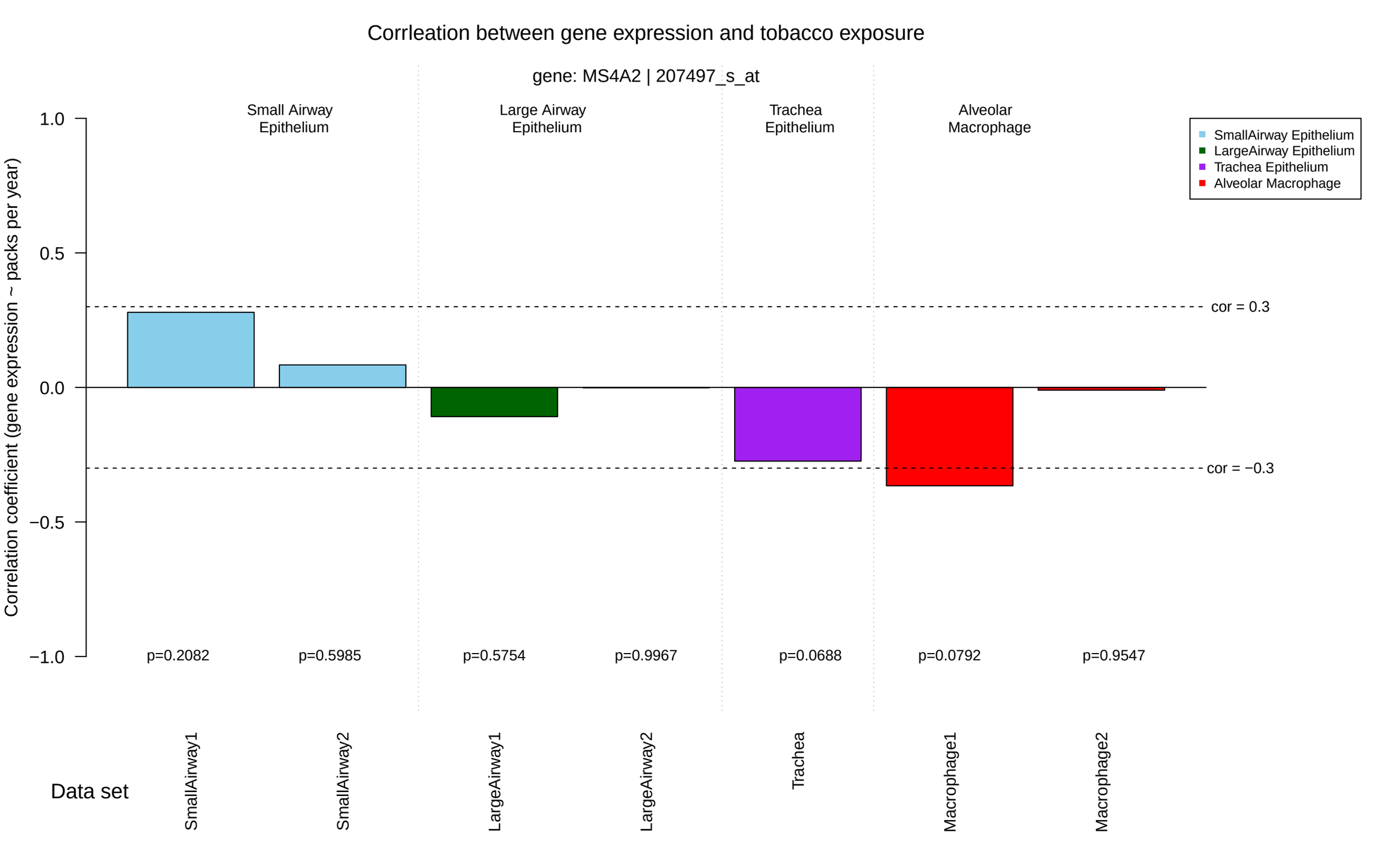

set: LargeAirway1 (p=0.9102)
MS4A2 | 207497_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5855)
MS4A2 | 207497_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0208)
MS4A2 | 207497_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1871)
MS4A2 | 207497_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9026)
MS4A2 | 207497_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MS4A2 | 207497_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2082
p=0.5985
p=0.5754
p=0.9967
p=0.0688
p=0.0792
p=0.9547
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2